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SUBJECT: COMPUTER VISION FOR IMAGE SEGMENTATION

SUBJECT-CODE: ITA0528

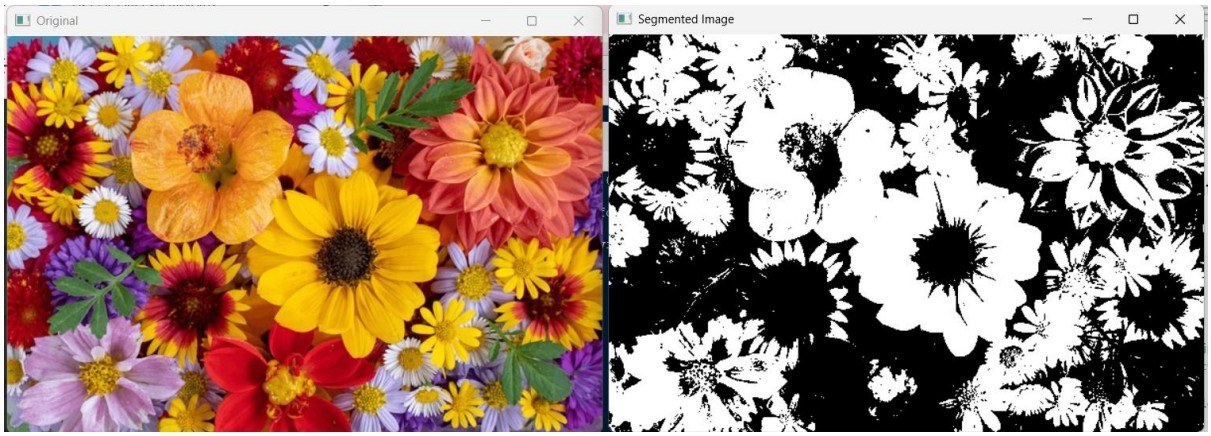
DATE: 12.03.2026

EXPERIEMNT – 31

Code:

```
import cv2
img = cv2.imread("sample.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
ret, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
cv2.imshow("Original", img)
cv2.imshow("Segmented Image", thresh)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



EXPERIMENT – 32

Code:

```
import cv2
import numpy as np
h = int(input("Enter height: "))
w = int(input("Enter width: "))
img = np.ones((h, w, 3), dtype=np.uint8) * 255
box_h = h // 10
box_w = w // 10
img[0:box_h, 0:box_w] = (0,0,0)       # Black
img[0:box_h, w-box_w:w] = (255,0,0)    # Blue
img[h-box_h:h, 0:box_w] = (0,255,0)    # Green
img[h-box_h:h, w-box_w:w] = (0,0,255)  # Red
cv2.imshow("Boxes", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

EXPERIMENT – 18

Code:

```
import cv2
import numpy as np
h = int(input("Enter height: "))
w = int(input("Enter width: "))
img = np.ones((h, w, 3), dtype=np.uint8) * 255
cv2.rectangle(img, (50,50), (200,150), (0,0,255), 3)
cv2.imshow("Rectangle", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

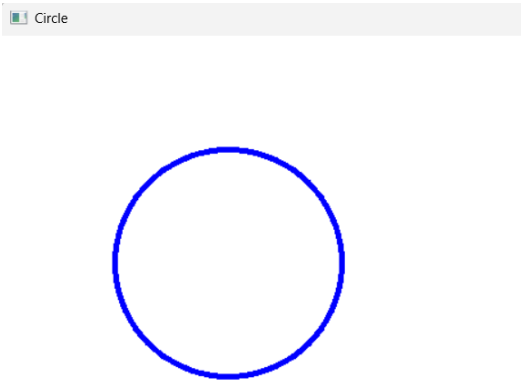
Output:

EXPERIMENT – 34

Code:

```
import cv2
import numpy as np
h = int(input("Enter height: "))
w = int(input("Enter width: "))
img = np.ones((h, w, 3), dtype=np.uint8) * 255
cv2.circle(img, (200,200), 100, (255,0,0), 3)
cv2.imshow("Circle", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

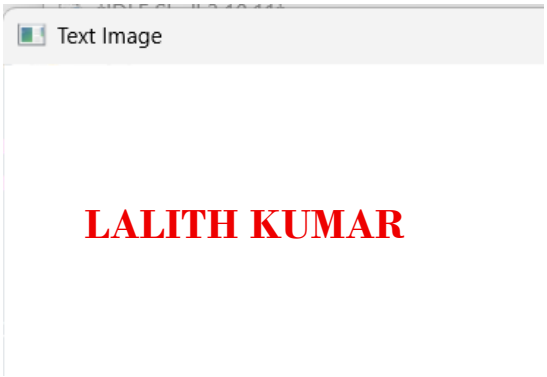


EXPERIMENT – 35

Code:

```
import cv2
import numpy as np
text = input("Enter text: ")
img = np.ones((400,600,3), dtype=np.uint8) * 255
cv2.putText(img, text, (50,200), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,0,255), 2)
cv2.imshow("Text Image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



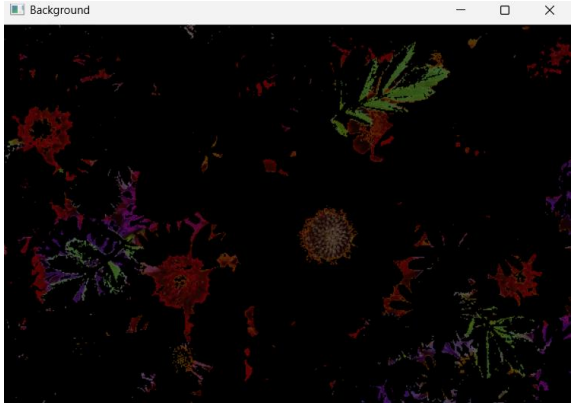
EXPERIEMNT – 36

Code:

```
import cv2
import numpy as np
def background_subtraction(image_path):
    img = cv2.imread(image_path)
    lower = np.array([0, 0, 0])
    upper = np.array([100, 100, 100])
    mask = cv2.inRange(img, lower, upper)
    result = cv2.bitwise_and(img, img, mask=mask)
    cv2.imshow("Original", img)
    cv2.imshow("Background", result)
```

```
    cv2.waitKey(0)
    cv2.destroyAllWindows()
background_subtraction("image.jpg")
```

Output:



EXPERIMENT – 37

Code:

```
import cv2
import numpy as np
def foreground_subtraction(image_path):
    img = cv2.imread(image_path)
    lower = np.array([100,100,100])
    upper = np.array([255,255,255])
    mask = cv2.inRange(img, lower, upper)
    result = cv2.bitwise_and(img, img, mask=mask)
    cv2.imshow("Original", img)
    cv2.imshow("Foreground", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
foreground_subtraction("image.jpg")
```

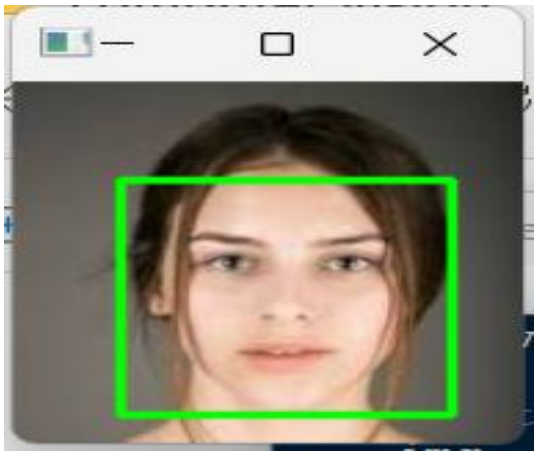
Output:

EXPERIMENT – 38

Code:

```
import cv2
def count_faces(image_path):
    img = cv2.imread(image_path)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
    faces = face.detectMultiScale(gray, 1.3, 5)
    print("Number of faces:", len(faces))
    for (x,y,w,h) in faces:
        cv2.rectangle(img,(x,y),(x+w,y+h),(0,255,0),2)
    cv2.imshow("Faces", img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
count_faces("face.jpg")
```

Output:



EXPERIMENT – 39

Code:

```
import cv2
def reverse_video(video_path):
    cap = cv2.VideoCapture(video_path)
    frames = []
    while True:
        ret, frame = cap.read()
        if not ret:
            break
        frames.append(frame)
for frame in reversed(frames):
        cv2.imshow("Reverse Video", frame)
        if cv2.waitKey(100) & 0xFF == 27:   # slow motion
            break
    cap.release()
    cv2.destroyAllWindows()
reverse_video("video.mp4")
```

Output:

20260312-0921-32.3512405.mp4

EXPERIMENT – 40

Code:

```
import cv2


def extract_text(video_path):
    cap = cv2.VideoCapture(video_path)
    while True:
        ret, frame = cap.read()
        if not ret:
            break
      gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
     edges = cv2.Canny(gray, 50, 150)
        contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
        for c in contours:
            x, y, w, h = cv2.boundingRect(c)
            if w > 50 and h > 20:
                cv2.rectangle(frame, (x,y), (x+w,y+h), (0,255,0), 2)
   cv2.imshow("Text Detection", frame)
   if cv2.waitKey(30) & 0xFF == 27:
            break
   cap.release()
    cv2.destroyAllWindows()
extract_text("video.mp4")
```

OUTPUT

20260312-0921-32.3512405.mp4